C Program to Print the Fibonacci Series

This beginner-friendly program prints the first n terms of the Fibonacci series. Each term after the first two is the sum of the previous two terms.

C Code

```
#include <stdio.h>

int main(void)
{
    int n;
    int first = 0, second = 1, next;

    printf("Enter the number of terms: ");
    scanf("%d", &n);

    if (n <= 0)
    {
        printf("Please enter a positive number.\n");
        return 1;
    }

    printf("Fibonacci series: ");

    for (int i = 0; i < n; i++)
    {
        printf("%d ", first);
        next = first + second;
        first = second;
        second = next;
    }

    printf("\n");
    return 0;
}
```

Sample Output

```
Enter the number of terms: 7
Fibonacci series: 0 1 1 2 3 5 8
```

How it works

1. The first two values are initialised as 0 and 1.

2. The loop prints the current first value.

3. next = first + second calculates the next Fibonacci number.

4. first = second and second = next move the values forward for the next iteration.

To compile with GCC: gcc fibonacci.c -o fibonacci

Run on Windows PowerShell: .\fibonacci.exe

Note: This example uses int, so very large terms can overflow. Use a smaller number of terms for this beginner exercise.